

Digital Assignment 1

1 $F(A, B, C, D) = \sum (0, 1, 3, 8, 10, 11, 13)$

A

CD \ AB	00	01	11	10
00	1	1	1	1
01				
11		1		
10	1		1	1

$$= B'C'D' + A'B'D' + ABC'D + ABC$$

Alternate

$$= ABC'D + A'B'C + B'CD + AB'D'$$

2 $F(X, Y, Z) = \sum m(0, 1, 6, 7)$

YZ \ X	00	01	11	10
0	1	1		
1			1	1

$$= A'B' + AB$$

3 $F(P, Q, R, S) = \prod M(1, 5, 6, 8, 11, 14)$
 $= \sum m(0, 2, 3, 4, 7, 9, 10, 12, 13, 15)$

RS \ PQ	00	01	11	10
00	1		1	1
01	1		1	
11		1		1
10	1	1	1	

$$= A'C'D' + B'C'D' + AC'D + AB'D + A'B'C + A'CD + ABCD$$

$$4 \quad F(A, B, C, D, E) = \sum m(0, 2, 4, 5, 8, 12, 13, 14, 20, 27, 28, 30, 31)$$

① Quine McCluskey Method

$F(A, B, C, D, E) = \sum m(0, 2, 4, 5, 8, 12, 13, 14, 20, 27, 28, 30, 31)$

	A	B	C	D	E
✓ 0	0	0	0	0	0
✓ 2	0	0	0	1	0
✓ 4	0	0	1	0	0
✓ 5	0	0	1	0	1
✓ 8	0	1	0	0	0
✓ 12	0	1	1	0	0
✓ 13	0	1	1	0	1
✓ 14	0	1	1	1	0
✓ 20	1	0	1	0	0
✓ 27	1	1	0	1	1
✓ 28	1	1	1	0	0
✓ 30	1	1	1	1	0
✓ 31	1	1	1	1	1

	A	B	C	D	E
0, 2	0	0	0	0	-
0, 4	0	0	1	0	-
4, 5	0	0	1	0	-
4, 12	0	0	1	0	-
4, 12, 20	0	0	1	0	-
8, 12	0	1	0	0	-
5, 13	0	1	1	0	-
12, 13	0	1	1	0	-
12, 14	0	1	1	1	-
20, 28	1	0	1	0	-
14, 30	1	1	1	0	-
28, 30	1	1	1	1	-
27, 31	1	1	1	1	-
30, 31	1	1	1	1	-

Prime Implicant Chart

	0, 2	4, 5	8, 12	13, 14	20, 28	27, 31	30, 31
A'B'C'E'	x						
ABDE							
ABCD							
A'D'E	x						
A'CD							
CD'E'							
BCE'							

$A'B'C'E' + ABDE + A'D'E + A'CD + CD'E' + BCE'$

$$5 \quad F(A, B, C, D, E) = \sum m(0, 1, 3, 7, 11, 13, 15)$$

	A	B	C	D	E
✓ 0	0	0	0	0	0
✓ 1	0	0	0	0	1
✓ 3	0	0	0	1	1
✓ 7	0	0	1	1	1
✓ 11	0	1	0	1	1
✓ 13	0	1	1	0	1
✓ 15	0	1	1	1	1

	A	B	C	D	E
0, 1	0	0	0	0	-
1, 3	0	0	0	1	-
3, 7	0	0	1	1	-
3, 11	0	1	0	1	-
13, 15	0	1	1	0	-
7, 15	0	1	1	1	-

	A	B	C	D	E
0, 1	0	0	0	0	-
1, 3	0	0	0	1	-
3, 7	0	0	1	1	-
3, 11	0	1	0	1	-
13, 15	0	1	1	0	-
7, 15	0	1	1	1	-

$\bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}DE + \bar{A}BCE$

	0	1	3	7	11	13	15
$\bar{A}\bar{B}\bar{C}\bar{D}$	x						
$\bar{A}BCE$							
$\bar{A}DE$							
$A'BCE$							

Q.2

$$i) (11011.0101)_2 = (9)_{10} = (9)_8 = (9)_{16}$$

$$16 \times 1 + 2 \times 1$$

$$11011$$

$$16 + 8 + 2 + 1$$

$$= 25$$

$$0101 = \frac{1}{2} + \frac{1}{16} = \frac{5}{16} = 0.3125$$

$$= (25.3125)_{10}$$

$$011011.010100$$

$$(33.24)_8$$

A	B	C	D	E	F
10	11	12	13	14	15

$$\frac{00011011}{(1B.5)_{16}}$$

$$ii) (CA75)_{16} = (9)_{10} = (9)_8 = (9)_2$$

$$CA75$$

$$(1100100101110101)_2$$

$$(144565)_8$$

$$2^0 + 2^2 + 2^4 + 2^5 + 2^6 + 2^8 + 2^{11} + 2^{14} + 2^{15}$$

$$1 + 2 + 16 + 32 + 64 + 256 + 2048 + 16384 + 32768$$

$$= 51573$$

$$iii) (9741.12)_{10} = (?)_8$$

9	7	4	1
1	4	8	7
0	2	4	3
1	1	2	1
1		6	0
0		3	0
0		1	5
0		7	6
0		3	8
0		1	9
1		9	
1		4	

$$= (10110000011001.0001)_2$$

$$= (26031.04)_8$$

0	1	2
0	0	2
0	0	4
0	0	8
1	1	6

$$iv) (107)_8 = (9)_{16}$$

$$\underline{00100011}_2 = (47)_8$$

$$v) (56088)_{10} = (710)_9$$

$$\begin{array}{r} 56088 \\ \underline{123} \end{array} = (456)_{10}$$

$$\begin{array}{r|l} & 456 \\ \hline 0 & 228 \\ \hline 0 & 114 \\ \hline 0 & 57 \\ \hline 1 & 28 \\ \hline 0 & 14 \\ \hline 0 & 7 \\ \hline 1 & 3 \\ \hline 1 & 1 \end{array}$$

$$\begin{array}{r} 111001000 \\ \hline (710)_8 \end{array}$$

$$vi) (ABDA)_{16} + (3AC9)_{16}$$

$$\begin{array}{r} 101010111101010 \\ + 0011101011001001 \\ \hline 1110011010100011 \\ \hline (E6A3)_{16} \end{array}$$

$$vii) (7531)_8 - (4442)_8$$

$$2' \text{ complement } \rightarrow (4442)_8 \rightarrow (10010010010)_2$$

$$\rightarrow 01101011101 + 1$$

$$110101110$$

$$(7531)_8 \rightarrow \begin{array}{r} 111111 \\ + (111101011001)_2 \\ 011011011110 \\ \hline 1011000110111 \end{array} \rightarrow (3067)_8$$

Q3 Binary to Gray

i) 11011101

→ 10110011

ii) 01110101
01001111

iii) 1111011
10000110

Q-4 Gray to binary

i) 11010101
10011001

ii) 01110101
01011001

iii) 10000110
11111011

Q-5 Number of one's counter ABCD=1001 xyz=00(e)

⇒

Tt	A	B	C	D	x	y	z
0	0	0	0	0	0	0	0
1	0	0	0	1	0	0	1
2	0	0	1	0	0	0	1
3	0	0	1	1	0	1	0
4	0	1	0	0	0	1	1
5	0	1	0	1	0	1	0
6	0	1	1	0	0	1	1
7	0	1	1	1	0	1	0
8	1	0	0	0	0	0	1
9	1	0	0	1	0	0	0
10	1	0	1	0	0	1	0
11	1	0	1	1	0	1	1
12	1	1	0	0	0	1	0
13	1	1	0	1	0	1	1
14	1	1	1	0	0	0	1
15	1	1	1	1	1	0	0

$$x = ABCD$$

$$Z(\text{Odd Parity}) = A \oplus B \oplus C \oplus D$$

$$y = 2s, 3s.$$

Quine McCluskey Method

$$Y = m \sum (3, 5, 6, 7, 9, 10, 11, 12, 13, 14)$$

	A	B	C	D		A	B	C	D
✓3	0	0	1	1	37	0	-	1	1
✓5	0	1	0	1	3, 11	1	-	1	1
✓6	0	1	1	0	5, 7	0	1	-	1
✓9	1	0	0	1	5, 13	-	1	0	1
✓10	1	0	1	0	6, 7	0	1	1	-
✓12	1	1	0	0	6, 14	-	1	1	0
✓7	0	1	1	1	9, 11	1	0	-	1
✓11	1	0	1	1	9, 13	1	-	0	1
✓13	1	1	0	1	10, 11	1	0	1	-
✓14	1	1	1	0	10, 14	1	-	1	0
					12, 13	1	1	0	-
					12, 14	1	1	-	0

(Cyclic Pattern)

$$= \underbrace{AB \oplus AC \oplus AD}_{\text{Appear 2 times}} \oplus \underbrace{BC \oplus BD \oplus CD}_{\text{Appear 3 times}}$$

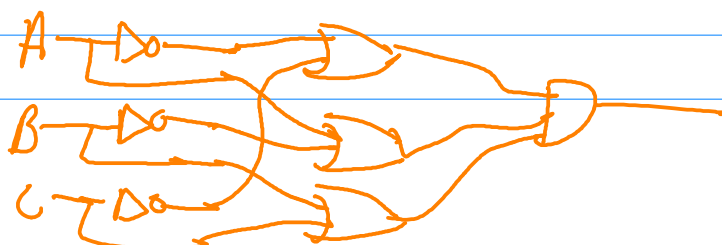
Q6 Circuit using decoder and logic gate.

a) $f(A, B, C) = \sum m(0, 2, 3, 4, 5, 7)$

⇒

B\C	00	01	11	10
A				
0	1	0	1	1
1	1	1	1	0

$$= A'C' + AB' + BC$$

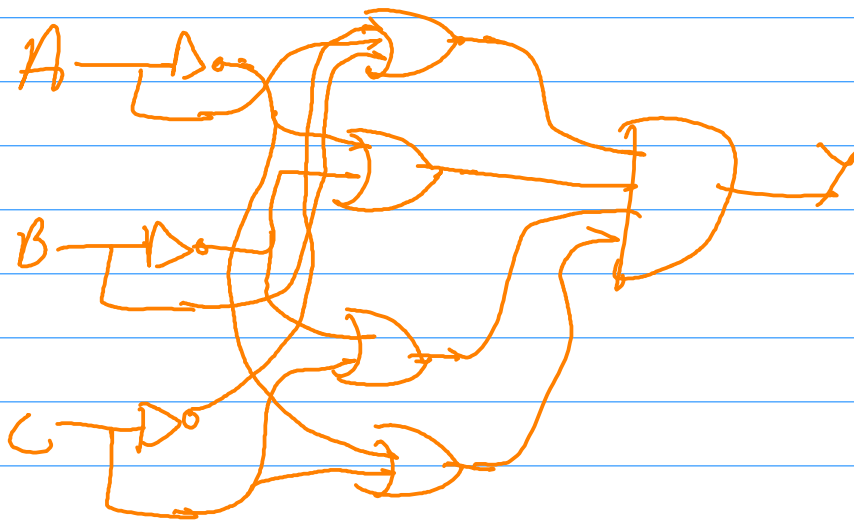


b) $f(A, B, C) = \pi M(2, 4, 7)$

$\Rightarrow f(A, B, C) = \sum m(0, 1, 3, 5, 6)$

A \ BC	BC			
	00	01	11	10
0	1	1	1	0
1	0	1	0	1

$Y = A'B' + B'C + A'C + ABC'$



c) $f(w, x, y, z) = \sum m(0, 4, 7, 12, 14)$

$\Rightarrow$

w \ yz	yz			
	00	01	11	10
00	1	0	0	0
01	1	0	1	0
11	1	0	0	1
10	0	0	0	0

$Y = wxz' + w'y'z' + w'xyz$

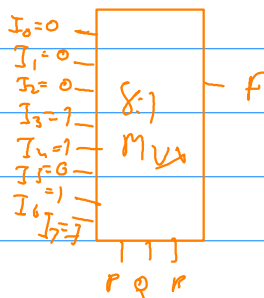
Circuit Implemented in Vivado

- Q.7 a) $F_1(P, Q, R) = \sum m(3, 4, 6, 7)$
 b) $F_2(X, Y, Z) = \pi M(1, 5, 6) \cdot D(4)$
 c) $F_3(a, b, c) = (a+b) \cdot (b'+c)$

a) i) 8:1 MUX

$$S_2 = P \quad S_1 = Q \quad S_0 = R$$

P	Q	R	m_{in}	F
0	0	0	0	0
0	0	1	1	0
0	1	0	2	0
0	1	1	3	1
1	0	0	4	1
1	0	1	5	0
1	1	0	6	1
1	1	1	7	1

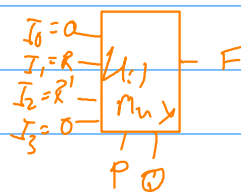


ii) 4:1 MUX (Input as first two selector)

$$S_1 = P \quad S_0 = Q$$

$R = ?$ (May be input)

P	Q	R	m_{in}	F
0	0	0	0	0 = 0
0	0	1	1	0
0	1	0	2	0 = R
0	1	1	3	1 = R
1	0	0	4	1 = R
1	0	1	5	0 = R
1	1	0	6	1
1	1	1	7	1



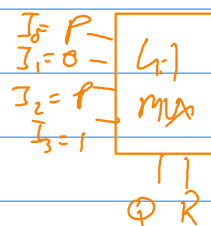
P	Q	F
0	0	0
0	1	R
1	0	R
1	1	1

ii) b) 4:1 MUX (Input as last 2 bit)

$P = \text{input ?}$

P	Q	R	m_{in}	F
0	0	0	0	0
0	0	1	1	0
0	1	0	2	0
0	1	1	3	1
1	0	0	4	1
1	0	1	5	0
1	1	0	6	1
1	1	1	7	1

Q	R	F
0	0	P
0	1	P
1	0	P
1	1	1

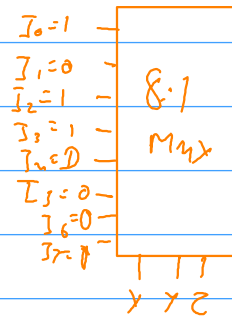


$$b) F^{(x,y,z)} = \sum m(1, 5, 6) D(4)$$

$$= \sum m(0, 2, 3, 7) \cdot D(4)$$

i) 8:1 Mux

S_2	S_1	S_0	M_m	F
0	0	0	0	1
0	0	1	1	0
0	1	0	2	1
0	1	1	3	1
1	0	0	4	D
1	0	1	5	0
1	1	0	6	0
1	1	1	7	1



ii) 4:1 Mux

a) first 2 as input

S_1	S_0	M_m	F
0	0	0	1
0	1	1	0
1	0	2	1
1	1	3	1

$$X = S_1, Y = S_0, Z = I_7$$

X	Y	F
0	0	1
0	1	1
1	0	0
1	1	2

$$F_2 = MUX_4(X, Y: Z', 1, 0, 2)$$

b = Last 2 as input

S_1	S_0	M_m	F
0	0	0	1
0	1	1	0
1	0	2	1
1	1	3	1

YZ	F
00	1
01	0
10	x'
11	1

$$F_2 = MUX_4(YZ: 1, 0, x', 1)$$

$$c) f_3(a, b, c) = ((a + b') \cdot (b' + c))'$$

$$= (a'b + bc')'$$

$$= b + a'c'$$

$$= b' + a.c$$

i) 8:1 Mux

S_2	S_1	S_0	M_m	F
0	0	0	0	1
0	0	1	1	1
0	1	0	2	0
0	1	1	3	0
1	0	0	4	1
1	0	1	5	1
1	1	0	6	0
1	1	1	7	1

$$f = MUX_8(a, b, c: 1, 1, 0, 0, 1, 1, 0, 1)$$

ii) $4:1$ Mux

a) first 2 as input

$S_1 = a, S_0 = b, C = I_q$

a	b	c	m _i	F
S_2	S_1	S_0		
0	0	0	0	1
0	0	1	1	1
0	1	0	2	0
0	1	1	3	0
1	0	0	4	1
1	0	1	5	1
1	1	0	6	0
1	1	1	7	1

a	b	F
0	0	1
0	1	0
1	0	1
1	1	1

$$F = \text{MUX}_4(a, b: 1, 0, 1, C)$$

b) last 2 as input

a	b	c	m _i	F
S_2	S_1	S_0		
0	0	0	0	1
0	0	1	1	1
0	1	0	2	0
0	1	1	3	0
1	0	0	4	1
1	0	1	5	1
1	1	0	6	0
1	1	1	7	1

$S_1 = b, S_0 = c, a = I_q$

b	c	F
0	0	1
0	1	1
1	0	0
1	1	0

$$F = \text{MUX}_4(b, c: 1, 1, 0, a)$$

Q-8

Given

x	y	z	m	F
0	0	0	0	0
0	0	1	1	1
0	1	0	2	1
0	1	1	3	0
1	0	0	4	1
1	0	1	5	0
1	1	0	6	0
1	1	1	7	1

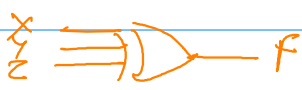
$$F = \sum m(1, 2, 4, 7)$$

Implement only using XOR

XOR $\Rightarrow$ Look for odd or even pattern

It is odd one's circuit among x, y, z

$$f = x \oplus y \oplus z$$



Q-5 skip

Q-10

$$F = AB' + BC + A'BC' + ABC$$

Only using NOR

$$= AB' + A'BC' + BC + ABC$$

$$AB' + A'BC' + BC(A + 1)$$

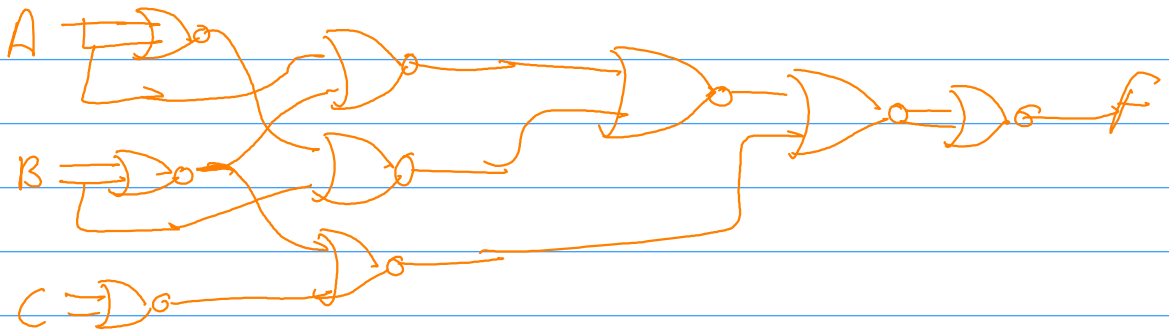
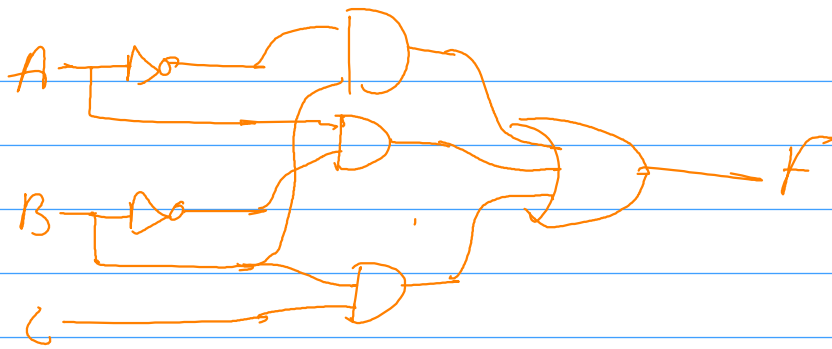
$$= AB' + A'BC' + BC$$

$$= AB' + B(A'C' + C)$$

$$= AB' + B(C + A')$$

$$AB' + BA' + BC$$

$$= AB' + A'B + BC$$



Q. 11

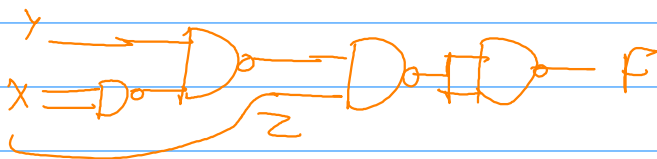
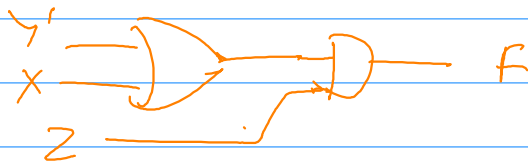
$$F = x'y'z + xy'z + xz$$

Only using NAND

$$= y'z(x' + x)$$

$$= y'z + xz$$

$$= z(y' + x)$$



Q. 12

Using 4:1 Mux = $F = \sum m(1, 2, 7, 8, 10, 12, 15, 16, 17, 18, 21, 24, 27, 29, 30)$

Max is 32 so 5 Variable A, B, C, D, E

We have S_1, S_0 as selector and

I_0, I_1, I_2, I_3 as Input

If $S_1 = A$ $S_0 = B$ $I_0, I_1, I_2, I_3 = C, D, E$

If $A, B = 0, 0$

Possible value of C, D, E cover 0, ①, ②, 3, 4, 5, 6, ⑦
 we want

AB	C	D	E
00	0	0	1
	0	1	0
	1	1	1

$$I_0 = C'D'E + C'DE' + CDE$$

If $A, B = 0, 1$

Possible value of C, D, E cover ⑧, 9, ⑩, 11, ⑫, 13, 14, ⑮
 we want

AB	C	D	E
01	0	0	0
	0	1	0
	1	0	0
	1	1	1

$$I_1 = C'D'E' + C'DE' + CDE' + CDE$$

If $AB = 1, 0$

Possible value of C, D, E cover ⑬, ⑭, ⑮, 16, 17, ⑱, 20, ⑲, 22, 23
 we want

AB	C	D	E
10	0	0	0
	0	0	1
	0	1	0
	1	0	1

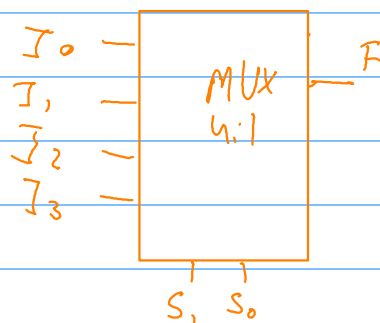
$$I_2 = C'D'E' + C'DE' + CDE' + CDE$$

If $AB = 1, 1$

Possible value of C, D, E cover ⑲, 24, 25, 26, ⑲, 28, ⑲, 30, 31

AB	C	D	E
11	0	0	0
	0	1	1
	1	0	1
	1	1	0

$$I_3 = C'D'E' + C'DE + CDE' + CDE$$



$$I_0 = C'D'E + C'DE' + CDE$$

$$I_1 = C'DE' + C'DE' + CDE' + CDE$$

$$I_2 = C'D'E' + C'DE + CDE' + CDE$$

$$I_3 = C'D'E' + C'DE + CDE' + CDE$$